

CHAPTER 20 SUMMARY

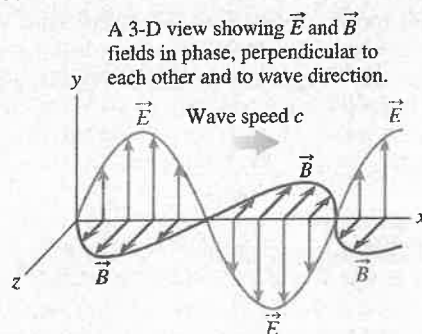
Electromagnetic Waves

(Section 20.1) An **electromagnetic wave** is a transverse wave, with oscillating electric and magnetic fields perpendicular to each other and to the direction of the wave's propagation. Electromagnetic waves include light, and the speed of all EM waves in vacuum is the **speed of light**, $c = 299,792,458 \text{ m/s}$. Electromagnetic waves carry energy and momentum and therefore exert **radiation pressure** on objects.

Speed of light in vacuum: $c = \sqrt{\frac{1}{\mu_0 \epsilon_0}} \approx 3.00 \times 10^8 \text{ m/s}$

Speed, wavelength, frequency: $c = \lambda f$

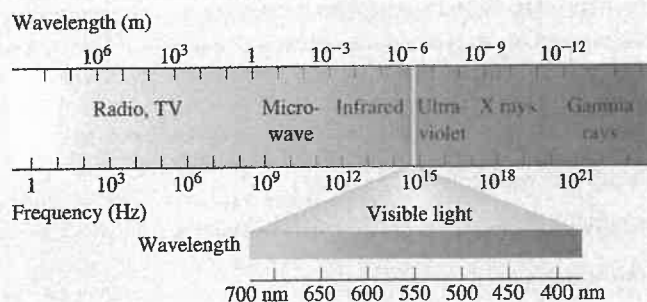
EM wave momentum p and energy U : $p = \frac{U}{c}$



The Electromagnetic Spectrum

(Section 20.2) The **electromagnetic spectrum** classifies EM waves according to frequency or wavelength. Radio waves have the longest waves, gamma rays the shortest. Visible light wavelengths range from 400 to 700 nm. Frequency is the inverse of wavelength—the shortest waves have the highest frequencies.

Wave type	Frequency (Hz)
Radio	$10^6 - 10^9$
Microwaves	$10^9 - 10^{12}$
Infrared	$10^{12} - 4.3 \times 10^{14}$
Visible light	$4.3 \times 10^{14} - 7.5 \times 10^{14}$
UV light	$7.5 \times 10^{14} - 10^{17}$
X rays	$10^{17} - 10^{20}$
Gamma	$> 10^{20}$



The Fundamental Speed c

(Section 20.3) The principle of relativity states that the laws of physics are the same in all inertial frames. That includes the fact that electromagnetic waves propagating in vacuum have speed c relative to all inertial frames. Einstein's theory of **special relativity** is based on this principle. Einstein's later theory of **general relativity** applies to accelerated reference frames.

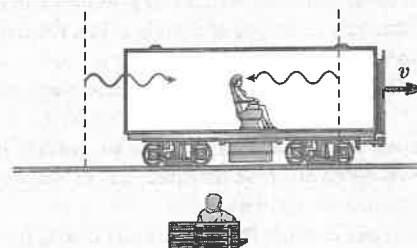
Relativity of Time and Space

(Section 20.4) Distinct events that are simultaneous in one inertial frame aren't necessarily simultaneous in another. **Time dilation** and **length contraction** describe how measures of time between events and lengths of objects differ among different inertial reference frames.

Relativistic factor γ : $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$

Time dilation: $\Delta t = \gamma \Delta t_0$

Length contraction: $d = \frac{d_0}{\gamma}$

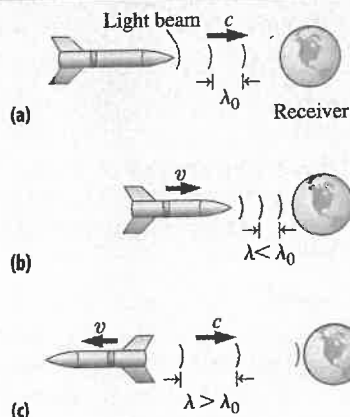


Relative Velocity and the Doppler Effect

(Section 20.5) Velocities transform from one reference frame to another using the relativistic velocity addition law. Although light's velocity never varies, its wavelength and frequency are Doppler shifted when source and receiver are in relative motion.

Relativistic velocity addition: $u_A = \frac{u_B + v}{1 + u_B v/c^2}$

Doppler effect: $\lambda = \sqrt{\frac{1 \mp v/c}{1 \pm v/c}} \lambda_0$



Relativistic Momentum and Energy

(Section 20.6) Relativistic momentum and relativistic kinetic energy grow without bound as an object's speed approaches c . An object also has rest energy, which is directly proportional to its mass. Rest and kinetic energies sum to the total relativistic energy.

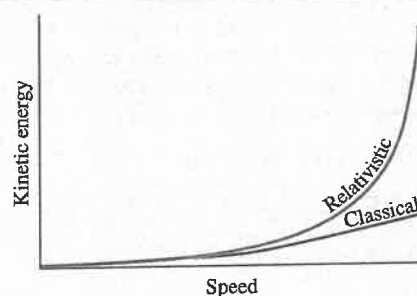
Relativistic momentum: $p = \gamma mv$

Relativistic kinetic energy: $K = (\gamma - 1)mc^2$

Rest energy: $E_0 = mc^2$

Total relativistic energy: $E = \gamma mc^2$

Energy-momentum relationship: $E^2 = p^2 c^2 + E_0^2$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled BIO are of biological or medical interest.

Conceptual Questions

- When light enters water, its frequency remains the same but its speed drops to about three-quarters of c . What happens to its wavelength?
- Discuss similarities and differences between light waves and sound waves.
- A torsion pendulum has a mirrored target on one side and a black target on the other. If the same light source strikes both the black and mirrored sides which way does the torsion pendulum rotate?
- You're in empty space, far from any sources of gravitational or other forces. If you turn on a flashlight and shine it in a fixed direction, what happens to you?
- Why would spacecraft propelled by "solar sails" need large sail areas?
- An electromagnetic wave is traveling from east to west. If its electric field oscillates in the north-south direction, in what direction does the magnetic field oscillate?
- The speed of light in water is about $0.75c$. If an electron is traveling at $0.90c$ in water, does this violate relativity's prohibition against speeds greater than c ?
- The "light clock" discussed in Section 20.4 seems rather contrived. Would other clocks keep time at the same rate?

- How would our lives be different if the speed of light were only 3×10^6 m/s? 30,000 m/s? 300 m/s? What if it were infinite?
- The star Sirius is 8.7 light years from Earth. Yet an astronaut can reach Sirius in less than 8.7 years (as measured by the astronaut) without traveling faster than light. How is this possible?
- An electron and proton have the same total energy E . Which has the larger kinetic energy? The larger momentum?
- The most distant galaxies recede from us with speeds close to c . How does this affect the (a) speed and (b) wavelength of light we receive from those galaxies?

Multiple-Choice Problems

- An electromagnetic wave with wavelength 570 nm in vacuum has frequency (a) 5.3×10^{11} Hz; (b) 5.3×10^{12} Hz; (c) 5.3×10^{13} Hz; (d) 5.3×10^{14} Hz.
- An electromagnetic wave whose peak electric field is 150 N/C has a peak magnetic field (a) 1.0×10^{-7} T; (b) 3.0×10^{-7} T; (c) 5.0×10^{-7} T; (d) 7.5×10^{-7} T.
- A 1360-nm electromagnetic wave is in what part of the electromagnetic spectrum? (a) Microwave; (b) ultraviolet; (c) visible; (d) infrared.

16. A 432-nm laser beam passes through a frequency-doubling device. What's its new wavelength? (a) 216 nm; (b) 432 nm; (c) 864 nm; (d) 1728 nm.
17. A rod with a proper length 1.0 m lies along the x -axis and is moving in the $+x$ -direction at $0.8c$ relative to you. You measure its length to be (a) 0.4 m; (b) 0.6 m; (c) 0.8 m; (d) 0.9 m.
18. A spaceship flies at $0.8c$ from Earth to Moon, a distance of 400,000 km. A clock on board the spaceship measures the trip time to be (a) 1.0 s; (b) 1.3 s; (c) 1.7 s; (d) 2.8 s.
19. A spaceship makes the 300-million-km round-trip journey to the Sun in just 20 min, as measured by clocks on Earth. How much time elapses on the spaceship's clock? (a) 20 min; (b) 17 min; (c) 14 min; (d) 11 min.
20. A spaceship approaching Earth at $0.5c$ fires a particle beam toward Earth. The particles move at $0.6c$ relative to the spaceship. At what speed does Earth receive the particles? (a) $0.75c$; (b) $0.8c$; (c) $0.85c$; (d) $0.90c$.
21. A spaceship approaching Earth at $0.5c$ fires a light beam toward Earth. At what speed does Earth receive the light beam? (a) $0.75c$; (b) $0.85c$; (c) $0.95c$; (d) c .
22. Two spacecraft approach Earth from opposite directions, each at $0.5c$ relative to Earth. Each spacecraft's speed as measured by the other is (a) $0.5c$; (b) $0.8c$; (c) $0.9c$; (d) c .
23. A spacecraft heading to the outer solar system at $0.20c$ sends radio signals back to Earth. The spacecraft transmits at a wavelength of 2.75 cm. What's the wavelength received on Earth? (a) 3.37 cm; (b) 3.75 cm; (c) 4.13 cm; (d) 5.50 cm.
24. The speed of a particle with kinetic energy equal to its rest energy is (a) $0.50c$; (b) $0.67c$; (c) $0.75c$; (d) $0.87c$.

Problems

Section 20.1 Electromagnetic Waves

25. ■ A helium-neon laser emits 632.8-nm-wavelength light. What's the frequency of this light, in vacuum?
26. ■ Find the wavelength ranges for (a) AM radio, with frequencies from 540 kHz to 1600 kHz, and (b) FM radio, with frequencies from 88 MHz to 108 MHz.
27. ■ If 546-nm laser light is passed through a frequency-doubling device, what's its new wavelength?
28. ■■ Show, given the SI units of μ_0 and ϵ_0 , that Equation 20.1 gives a result whose units are m/s.
29. ■■ Light travels at about $0.75c$ in water. What's the wavelength in water of light whose wavelength in vacuum is 624 nm?
30. ■■ The height of a typical AM radio station's tower is one-fourth of the radio wavelength. (a) How high should the tower be for a radio station broadcasting at 710 kHz? (b) If another station uses a higher frequency, should its tower be taller or shorter?
31. ■■ (a) Find the peak magnetic field in an electromagnetic wave whose peak electric field is 150 N/C. (b) Find the peak electric field in an electromagnetic wave whose peak magnetic field is 25 nT.
32. ■■ Solar radiation reaches Earth at a rate of about 1400 W/m^2 . If all this energy were absorbed, what would be the average force exerted by radiation pressure on a square meter oriented at right angles to the sunlight?
33. ■■ A 5.0-mW laser beam has a circular cross section with diameter 0.60 mm. What is the maximum radiation pressure that this laser can exert on (a) a mirrored surface and (b) a black surface?

Section 20.2 The Electromagnetic Spectrum

34. ■ For electromagnetic waves with each of these frequencies, find the wavelength in vacuum. In what region of the electromagnetic spectrum does each lie? (a) $9.4 \times 10^{12} \text{ Hz}$; (b) $6.2 \times 10^{14} \text{ Hz}$; (c) $2.9 \times 10^{15} \text{ Hz}$; (d) $5.0 \times 10^{17} \text{ Hz}$.
35. ■ For electromagnetic waves with each of these wavelengths (in vacuum), find the frequency. In what region of the electromagnetic spectrum does each lie? (a) 2.0 km; (b) 2.0 m; (c) 2.0 mm; (d) $2.0 \mu\text{m}$; (e) 2.0 nm.
36. ■ A wireless computer network uses microwaves at 5.0 GHz. What's the corresponding wavelength (in air, where λ is negligibly different than in vacuum)?
37. ■ Referring to Figure 20.6, find the range of frequencies in the microwave part of the electromagnetic spectrum.
38. ■ What is the frequency of the 12-cm microwaves used in a microwave oven?
39. ■ Antennas for radio transmission and reception are particularly effective if they're about a half-wavelength long. What's the appropriate length for an FM antenna, taking 100 MHz as a typical FM frequency? (You may have used such an antenna with your FM receiver.)
40. ■■ Suppose you set up a standing wave pattern using light at the red end of the visible spectrum. What's the distance between nodes in this standing wave?

Section 20.3 The Fundamental Speed c

41. ■ An electron is moving at $0.99c$ directly toward a light source. In the electron's reference frame, how fast is the light approaching?
42. ■■■ A boat moves at 6.00 m/s relative to the water. Find the boat's speed relative to shore when it's traveling (a) downstream and (b) upstream in a river with a 2.0-m/s current. (c) The boat travels 100 m downstream and then 100 m upstream, returning to its original point. Find the time for the round trip, and compare this time with the round-trip time if there were no current. (You can neglect relativity at these slow speeds.)
43. ■■■ For the boat and river of the preceding problem, compute the travel time for a round trip on a 100-m path perpendicular to the current, again assuming the boat returns to its starting point. Compare the result with that of the preceding problem. (Michelson's instrument sent light beams along two such perpendicular paths, one going with the Earth's velocity and one perpendicular to it.)

Section 20.4 Relativity of Time and Space

44. ■ A rod with a proper length of 10.0 m lies along the x -axis. If the rod is traveling in the $+x$ -direction, find the rod's measured length when its speed is (a) $0.1c$; (b) $0.5c$; (c) $0.9c$; (d) $0.99c$.
45. ■■ A spaceship flies from Earth to Moon, a distance of 400,000 km. Find the elapsed time for this trip both on Earth and on a clock on board the spaceship when the spaceship's speed is (a) $0.01c$; (b) $0.3c$; (c) $0.75c$; (d) $0.99c$.
46. ■■ Our galaxy's diameter is about 10^5 light years. How much time does it take an electron moving at $0.9999c$ to cross the galaxy, measured in the reference frame of (a) the galaxy and (b) the electron?
47. ■■■ Suppose the light clock in Figure 20.11 is 1.5 m high and is moving at $0.5c$. (a) Find the elapsed time for the light's round trip as measured by (a) Sharon on the train and (b) Sam on the ground. (c) What distance does the light travel, as measured by Sam? (d) Use your answers to parts (b) and (c) to compute the speed of the light, as measured by Sam.
48. ■■ You want to gain a year on your earthbound peers, traveling fast enough on a round trip that when 10 years elapse for them, only 9 years elapse for you. What speed is required?

49. ■■ What speed will shorten the length of a round-trip journey by 1%, compared with its duration measured in an inertial reference frame?
50. ■■ In the future, speeding tickets for spaceships will be issued based on the spaceship's measured length. The Neptune 3000 spaceship has proper length 75 m. If the posted speed limit is $0.6c$, will the Neptune 3000 spaceship be ticketed if space police measure its length as 65 m?
51. ■■ A future spaceship makes the round-trip journey to the Moon in just 1 min, as measured by clocks on Earth. How much time elapses on a clock aboard the spaceship? (Use the average Earth-Moon distance listed in Appendix E.)
52. ■ Find the speed of a spacecraft if its clock, relative to clocks on Earth, runs at a rate (a) 90% as fast; (b) 60% as fast; (c) 10% as fast.
53. ■ A linear accelerator can accelerate subatomic particles to close to the speed of light. A proton travels through a 1-km section of the accelerator with a constant speed. Find the proton's speed if this section is only 1.0 m long in the proton's reference frame.
54. ■■ Sirius is a large blue star just 8.7 light years from Earth. Find the speed required for a one-way trip to Sirius that takes, as measured on board the spacecraft, (a) 40 years; (b) 20 years; (c) 5 years.
55. ■■ A subatomic particle has a 480-ns lifetime in its own rest frame. If it moves through the lab at $0.980c$, how far does it travel before decaying, as measured in the lab?
56. ■■ Spaceship A has proper length 100 m, and ship B has a proper length 50 m. They pass an observer on Earth, both moving in the same direction but with different speeds. The observer measures both ships' lengths to be 42 m. How fast is each moving?
57. ■■■ A relativistic train with proper length 80 m passes through a tunnel with a proper length 75 m. (a) If the train's speed is $0.45c$, is the train ever completely inside the tunnel? (b) At this speed, how much time elapses for a cross-country trip of 5000 km, as measured on the train?
58. ■■■ A spaceship travels at $0.40c$ relative to an observer. How much time does the spaceship's clock lose each day, compared with the observer's? Does the observer's clock gain this much time compared with the spaceship's? Explain.

Section 20.5 Relative Velocity and the Doppler Effect

59. ■ A spaceship approaching Earth at $0.73c$ fires a particle beam toward Earth, with speed $0.88c$ relative to the ship. At what speed does Earth receive the particles?
60. ■ Repeat the preceding problem if the ship is moving away from Earth at $0.73c$ but still fires the particle beam toward Earth.
61. ■■ Two spaceships approach Earth from opposite directions, each traveling at $0.7c$ relative to Earth. How fast is each moving, as measured by the other?
62. ■ An astronaut returning to Earth after a long trip is cruising earthward at 6.95×10^7 m/s. She launches a package toward home, at a speed of 1.50×10^7 m/s relative to her spacecraft. At what speed does the package reach Earth?
63. ■ A source emitting 540-nm light moves toward a receiver. Find the received wavelength for source speeds (a) $0.01c$; (b) $0.10c$; (c) $0.50c$; (d) $0.90c$; (e) $0.99c$.
64. ■ Repeat the preceding problem if the source moves away from the receiver.
65. ■■ A proton is moving at $0.25c$ to the right, as viewed in the laboratory. Find the velocity (relative to the lab) of another reference frame in which the proton is moving at $0.25c$ to the left.
66. ■■■ Three galaxies lie along a line. The outer two galaxies are receding at $0.25c$ relative to the middle one. (a) Determine the speed of each outer galaxy in the reference frame of the other outer galaxy. (b) If each galaxy emits visible light with wavelength 550 nm, at what wavelength does one of the outer galaxies receive light from the other two?
67. ■■ A light source moves toward a receiver with speed v . If the source emits light with a frequency f_0 , what is the frequency of the received signals? *Hint:* Remember that $\lambda f = c$ for electromagnetic waves. Evaluate numerically for a source of 200-nm ultraviolet light moving toward an observer at $0.4c$.
68. ■■ Repeat the preceding problem if the source is moving away from the observer.
69. ■■ A space probe is leaving the solar system at $0.15c$, moving directly away from Earth. It emits radio pulses at the rate of 40 per minute. At what rate are the pulses received at Earth?
70. ■■ What relative speed between the sender and receiver is required to shift light from one end of the visible spectrum to the other (400 nm to 700 nm)?
71. ■■ Particle accelerators often employ colliding-beam technology, in which fast particles approach a target chamber from opposite directions. If each beam approaches the chamber at $0.990c$, what's each beam's speed relative to the other?

Section 20.6 Relativistic Momentum and Energy

72. ■■ Find the momentum, kinetic energy, and total energy of a proton with speed (a) $0.01c$; (b) $0.10c$; (c) $0.50c$; (d) $0.99c$.
73. ■■ Find the momentum, kinetic energy, and total energy of an electron with speed (a) $0.01c$; (b) $0.10c$; (c) $0.50c$; (d) $0.99c$.
74. ■■ A proton is accelerated from rest through a potential difference of 100 MV. Find the proton's kinetic energy and rest energy. What's the ratio of kinetic to rest energy?
75. ■■ A positron is the antiparticle of the electron, with the same mass but charge $+e$. Find the minimum energy required to produce an electron-positron pair.
76. ■■ PET Imaging. Positron-emission tomography (PET) images body tissues by detecting gamma rays produced when electrons and positrons annihilate. (The positrons come from short-lived radioisotopes introduced into the patient.) What's the energy of such gamma rays? See the preceding problem.
77. ■■ A particle's kinetic energy is equal to its rest energy. What's its speed?
78. ■■ (a) Find the speed of an electron with a kinetic energy 4.80×10^{-13} J. (b) What's the speed of a proton with the same kinetic energy?
79. ■■■ (a) Find the speed required for the classical and relativistic values of (a) momentum and (b) kinetic energy to differ by 1%.
80. ■■ An alpha particle (helium nucleus) has total energy 4.1 GeV. (a) What's its speed? (b) What's the speed of an electron with the same total energy?
81. ■■ A high-energy electron has $\gamma = 100$. Find (a) its momentum, (b) its total energy, and (c) the ratio of its total energy to its rest energy.
82. ■■ An atom moving at $0.890c$ has a momentum 1.15×10^{-16} kg · m/s. Find its mass.
83. ■■ A proton's total energy is ten times its rest energy. How fast is it moving?

General Problems

84. ■■ Find (a) the relativistic factor γ and (b) the speed of protons accelerated to energy 7 TeV in the Large Hadron Collider.
85. ■■ What power laser beam would be needed to suspend a 2.7-g ping-pong ball against Earth's gravity? The ball is painted black so it absorbs the entire beam.

86. ■■■ How much ice must melt at 0°C in order to gain 1 g of mass? What's the source of this mass? (The latent heat of ice melting at this temperature is 333 kJ/kg .)
87. ■■ A stretched spring has greater energy and therefore greater mass than an unstretched spring. What's the mass increase when you stretch a spring with $k = 480\text{ N/m}$ by 20 cm?
88. ■■■ A spaceship heads away from Earth at $0.20c$. The ship beams 400-nm light toward Earth. The light strikes a mirror on Earth and is reflected back to the ship. What is the wavelength of light received by the ship? *Hint:* Find the wavelength of light received at Earth, and then consider that the mirror effectively retransmits light of that wavelength.
89. ■■ Imagine a "moving walkway" like you find in airports, except that this one goes super fast: $0.75c$, relative to the airport. Suppose you put an identical walkway on top of this one. How fast are passengers on the second walkway moving relative to the airport?
90. ■■■ You embark on a round-trip journey to a star 10 light years away, as measured in Earth's reference frame. You travel at a constant speed outbound, turn around rapidly, and travel back at constant speed. If the round trip takes 15 years by your clock, (a) how fast do you travel and (b) what's the round-trip time as measured on Earth?
91. ■■ Hydrogen atoms throughout the universe emit ultraviolet radiation with wavelength 121.6 nm. How fast must a galaxy be receding from us in order that we receive this radiation as visible light with wavelength 650 nm?
92. ■■ The most distant galaxies observed have redshifts of about 7, meaning the wavelength of light we receive is seven times longer

than what was emitted at the galaxy. How fast is a redshift-7 galaxy moving, and is it receding or approaching?

93. ■■ Using the momentum and energy found in Example 20.11, show that Equation 20.14 gives the rest energy found in that example.
94. ■■■ The expression $(1 + x)^n$ can be approximated as $1 + nx$, provided $|x| \ll 1$. Here the exponent n does not have to be an integer. Use this approximation to show that Equation 20.11's expression for relativistic kinetic energy reduces to $K = \frac{1}{2}mv^2$ for $v \ll c$.

Answers to Chapter Questions

Answer to Chapter-Opening Question

Nearly all our information about the universe beyond Earth comes from electromagnetic waves, with astronomers now able to access a spectrum that ranges from radio waves (received by the radio telescopes shown in the chapter-opening photo) to gamma rays.

Answers to GOT IT? Questions

- Section 20.1 (b) $1.05\text{ }\mu\text{m}$ < (d) 780 nm < (a) 550 nm < (c) 434 nm
- Section 20.2 (c) 10^{14} Hz
- Section 20.4 (a) Shannon measures a shorter time than Chris.
- Section 20.5 (c) More than $0.6c$ but less than c
- Section 20.6 (a) larger for the proton